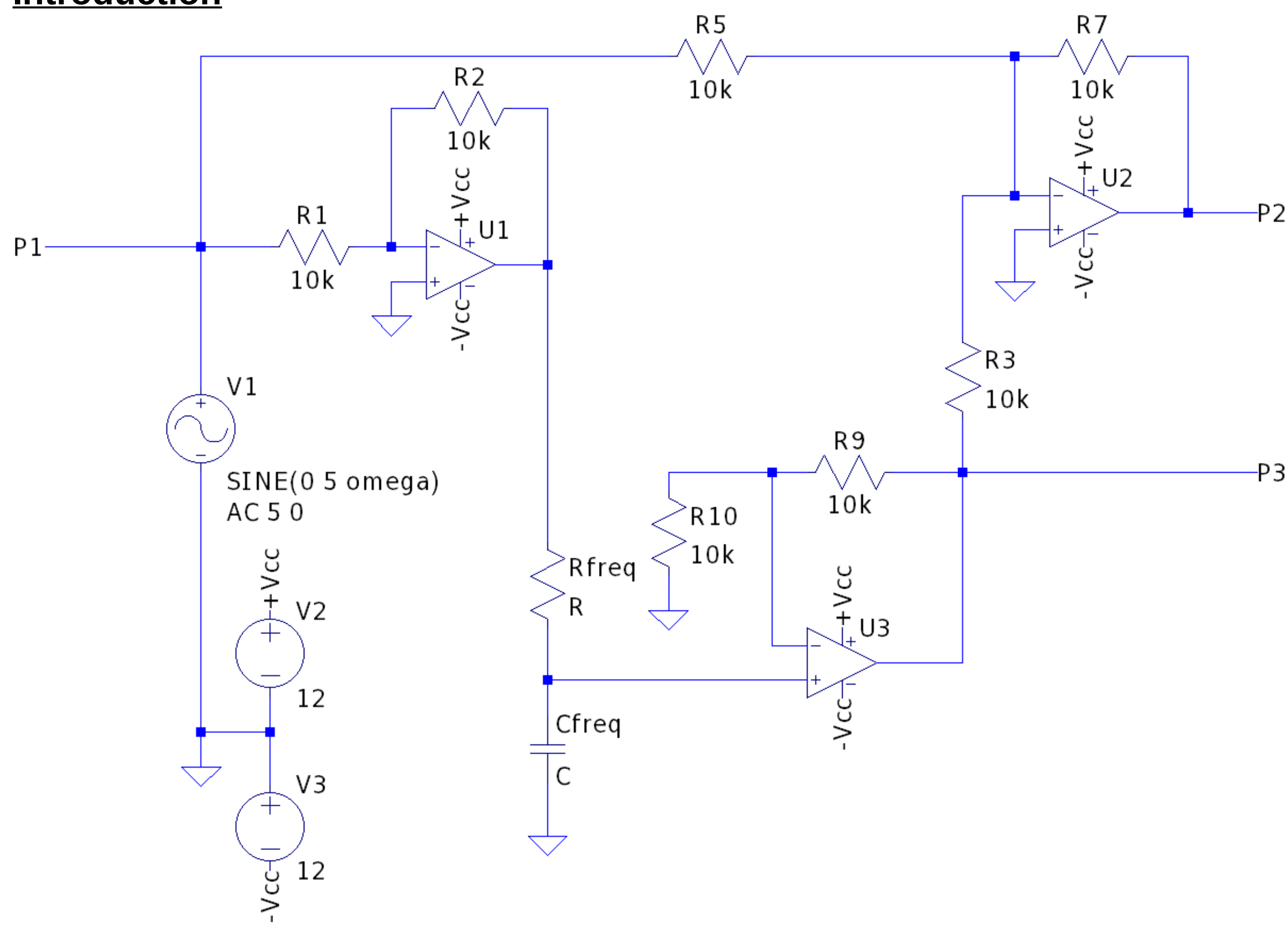
Lab 11 - Simulated 3-Phase AC Circuit

Objective

The goal of this lab is to further familiarize students with 3-phase circuits as well as using op-amps via a simulated circuit.

Introduction



We can simulate a 3-phase AC circuit using the 3 op-amp circuit shown above. By adjusting Rfreq and Cfreq, we can set the frequency of the AC signal such that P1, P2, and P3 are at the same voltage, but 120° out of phase with respect to each. How does this circuit accomplish this shift?

We start with the input (P1), which is our “phase 1”. That input gets sent to two op-amps. In the first op-amp (U1), as that’s a simple unity inverter, the output is 180° shifted with respect to P1. By sending that output through an RC, we can create an additional phase shift by choosing the appropriate R and C values. As we want P3 to be 240° shifted from P1, we need to add another 60° shift to our already shifted 180° signal. Thus, we need to pick R and

C such that $\tan^{-1}(R/Z_c) = 60°$, where Z_c is $1/\omega C$, or $RC = \tan(60°)/\omega = \sqrt{3}/\omega$. Thus, if our input signal is 1kHz and we use a 100nF capacitor, we need to choose R to be ~2.76kΩ. By sending this signal through a non-inverting 2X gain, op-amp (U3), we get P3 shifted by 240° with respect to P1 (we set the gain to 2X in order to overcome the signal loss from the RC configuration).

The final op-amp (U2) is set up as a summer of the input (P1) and the output of U3 (P3). Since P1 is at 0° and P3 is at 240°, the resulting sum has ashift of -60°, or 300°. By running this signal through a unity inverter, we shift the signal another 180°, resulting in a shift of 480°, or 480°-360° = 120°. Thus, P2 will be 120° shifted with respect to P1. Thus, by using a set of three op-amps and an RC phase shifter, we can simulate a 3-phase AC signal.

Procedure

Parts and equipment needed:

- 3 resistors between 1kΩ and 10kΩ
- 3 capacitors between 100nF and 1μF
- A selection of colored 22 gauge connection wires
- XK700 Trainer
- Oscilloscope

Build the circuit shown in the introduction. Note, you will actually build 9 circuits, with each one using a different RC combination. For each combo, calculate the frequency for which the circuit will produce a proper 3-phase AC signal. Set your signal generator close to that frequency and measure the voltage and phase shift (with respect to P1) at P3. Adjust your signal generator until the voltage at P3 is the same as P1 and record that frequency. Measure the phase shift at P2.

Example Table of Results format

R	C	f (calculated)	f (measured)	ϕ (P1)	ϕ (P2)

Lab Report

- Title page
- Procedure: Write in your own words what you did in lab
- Data and Results: Provide a table with your results
- Discussion:
  - Did the measured values match your calculated values? If not, discuss the causes for the discrepancy.
  - How might this circuit be used in application?